

Question				Marking details				Marks available													
								AO1	AO2	AO3	Total	Maths	Prac								
4	(a)			<table><tr><td>Radio waves</td><td>Microwaves</td><td>Infra-red</td><td>Visible</td></tr><tr><td>$4 \times 10^2 \text{ m}$</td><td>$2 \times 10^{-2} \text{ m}$</td><td>$3 \times 10^{-5} \text{ m}$</td><td>$5 \times 10^{-7} \text{ m}$</td></tr></table>				Radio waves	Microwaves	Infra-red	Visible	$4 \times 10^2 \text{ m}$	$2 \times 10^{-2} \text{ m}$	$3 \times 10^{-5} \text{ m}$	$5 \times 10^{-7} \text{ m}$		2		2		
				Radio waves	Microwaves	Infra-red	Visible														
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All correct award 2 marks 2 or 3 correct award 1 mark 1 correct 0 marks																					
	(b)	(i)		UV, X- <u>rays</u> , gamma [rays] – all 3 needed				1			1										
		(ii)		[Radiation that] has high energy / interacts with atoms / removes electrons from atoms / damages <u>cells</u> Don't accept causes cancer / mutations				1			1										
	(c)			Transfer energy / travel at the same speed [in a vacuum] / can travel through empty space Don't accept transmit information				1			1										

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	(d)			Correct substitution and rearrangement into the speed formula (1) $t = \frac{7.2 \times 10^7}{3 \times 10^8} = 0.24 \text{ [s]} \text{ (1)}$ $t = \frac{6.0 \times 10^6}{2 \times 10^8} = 0.03 \text{ [s]} \text{ (1)}$ So disagree Conclusion must be present to award all 3 marks Alternative: Speed of the signal to the satellite is 1.5 times faster than in the cable (1) The distance travelled to the satellite and back is 12 times greater than the cable (2) So disagree Conclusion must be present to award all 3 marks Award a maximum of 2 marks for answers of 0.48 [s] and 0.06 [s]			3	3	3	
				Question 4 total	3	2	3	8	3	0